

Smart Parking System Using IR Sensors and Servo-Controlled Toll Gate and Duration-Based Fare Estimation

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Abstract—This paper presents an automated smart parking system using IR obstacle sensors and an Arduino Uno microcontroller. The system detects vehicle entry/exit via IR sensors, monitors parking spot occupancy (two slots), and controls a servo-operated toll gate. An LCD display guides users by indicating vacant slots during entry and calculates parking duration during exit for fare estimation. The prototype demonstrates real-time slot detection and responds to vehicle movements. This low-cost solution reduces manual intervention and optimizes urban parking management.

Keywords — Smart parking; IR sensors; Arduino; servo motor; LCD display

I. INTRODUCTION

Urban areas face significant challenges in managing parking due to the rising number of vehicles and limited parking space. Traditional parking systems often require manual supervision, leading to delays, confusion, and inefficient use of available spots. To address these issues, this project presents a compact, Arduino-based smart parking system utilizing four IR sensors, a servo motor, and an LCD display. The system uses two IR sensors to detect vehicle entry and exit, while the remaining sensors monitor parking slot occupancy. A servo motor operates the toll gate based on vehicle movements, and the LCD provides real-time information, such as available parking spots during entry and parking duration for fare estimation upon exit. This low-cost solution automates the parking process, enhances the user experience, and optimizes parking space usage, making it ideal for small-scale environments like college campuses or residential complexes[1].

II. MEASUREMENT TECHNIQUE

The smart parking system employs IR (Infrared) obstacle sensors for accurate, real-time vehicle detection. Each sensor consists of a transmitter that emits infrared light and a receiver that detects the reflected beam when obstructed. The system uses four IR sensors: two placed at the entry and exit points to detect vehicle movement, and two placed at individual parking spots to monitor occupancy. When a vehicle crosses the entry sensor, the system logs the timestamp and opens a servo-controlled toll gate. Upon exit, the system logs the timestamp again to calculate the total parking duration. The slot sensors

continuously detect vehicle presence, updating the parking status in real time.

These sensor outputs are processed by an Arduino Uno, which controls the servo motor and LCD display. The servo motor operates the gate in response to vehicle movement, while the LCD shows available parking slots during entry and the total parking duration at exit for fare calculation. This IR-based system is non-contact, cost-effective, and efficient, offering high responsiveness and reliability for automated parking management

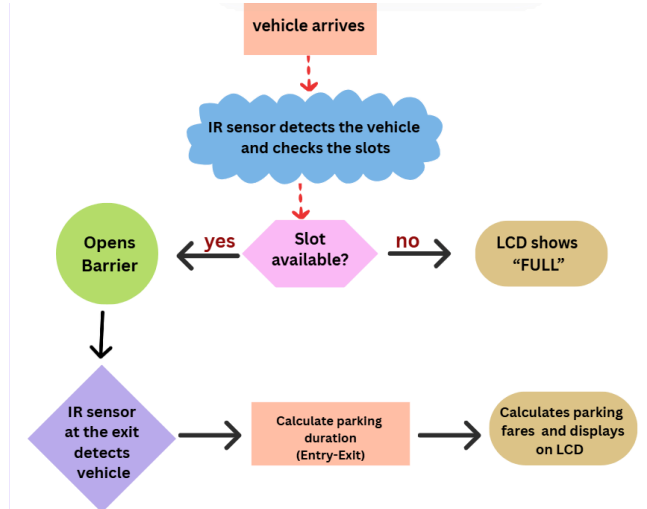


Fig. 1: Schematic of the prototype

III. DESCRIPTION OF THE PROTOTYPE

The smart parking system prototype, shown in Fig. 1, is designed using various components: an Arduino Uno microcontroller, IR obstacle detection sensors, a 16x2 LCD display, a servo motor, jumper wires, and a foam-based structural model. This compact and portable system simulates a miniature parking structure that can detect vehicle entry, slot availability, and the status of the parking lot.

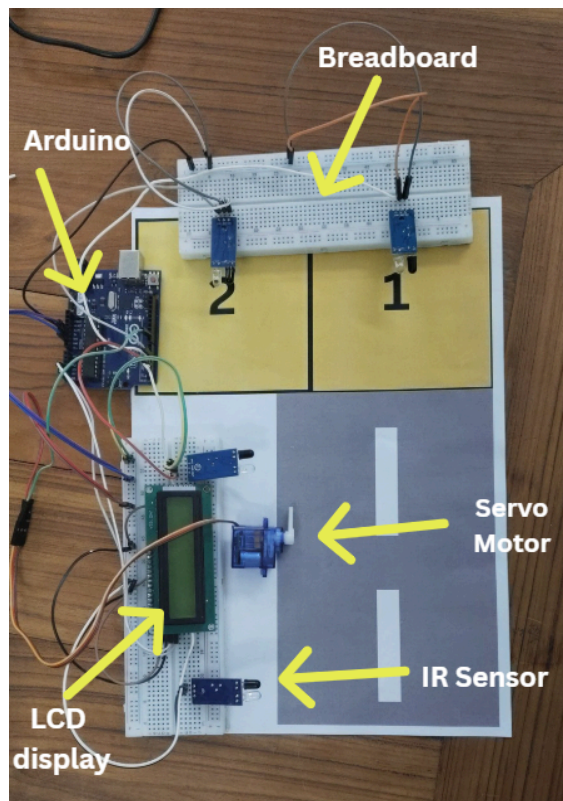


Fig. 2: Photograph of the Prototype of Smart parking system.

A. Vehicle Entry Detection:

When a vehicle approaches, the entry IR sensor detects it and signals the Arduino, which commands the servo motor to open the gate to 90°. The LCD displays “Gate Open – Welcome” along with the number of free slots.

B. Slot Occupancy Detection:

As the vehicle parks, the corresponding IR sensor marks the slot as “Occupied.” The Arduino updates the LCD in real time to reflect current slot availability..

C. Parking Full Notification:

If all slots are occupied, the LCD shows a “Full” message, and the entry gate remains closed to prevent additional vehicles from entering..

D. Vehicle Exit Detection:

When a vehicle exits, the exit IR sensor signals the Arduino to open the exit gate and updates the LCD, marking the slot as “Vacant.”

E. Time and Data Logging:

The system records entry and exit times in hh:mm:ss format to calculate the parking duration. This information is shown on the LCD or stored for later use..

IV. EXPERIMENTAL RESULTS

Controlled experiments evaluated the performance and reliability of the miniature parking system prototype in a simulated environment. The setup included a toy car-based

small-scale parking lot with two slots, each equipped with IR proximity sensors connected to an Arduino Uno. These sensors detected car presence, allowing the Arduino to update slot status in real time. A 16x2 LCD displayed availability, and the system was powered by a constant 5V supply.

Cars were repeatedly inserted and removed to test sensor accuracy and responsiveness. Each slot underwent over 30 occupancy cycles under normal indoor lighting. The LED indicators and LCD responded accurately with minimal lag or false detections, supporting real-time monitoring. In a two-hour test, the LCD showed stable performance with no freezing or flickering. Powered by a 10000mAh power bank, the system operated for over five hours, confirming its power efficiency and suitability for portable or remote use.

Minor issues included occasional sensor misreads under direct sunlight resolved with shading and loose breadboard connections causing momentary signal loss. These are common prototype limitations and can be addressed using a soldered PCB in future versions.

Overall, the prototype demonstrated reliable occupancy detection, consistent display updates, and efficient power usage, suggesting strong potential as a scalable foundation for a full-scale smart parking system .

V. CONCLUSIONS

A miniature smart parking system was successfully developed using Arduino, IR sensors, and toy car models to simulate real-time vehicle detection and slot monitoring. The prototype is low-cost, compact, and effectively demonstrates the core functionality of a smart parking solution. Its automated operation and simple design make it ideal for learning and testing. Moving forward, the system could be scaled up by integrating more parking slots, cloud-based data management, and mobile applications for real-time monitoring and user interaction[2]. This project serves as a foundational step towards building a fully functional smart parking system suitable for real-world use, optimizing urban parking and reducing manual intervention.

ACKNOWLEDGEMENT

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